C60 regularisation: minimum-shifted zero-fill map

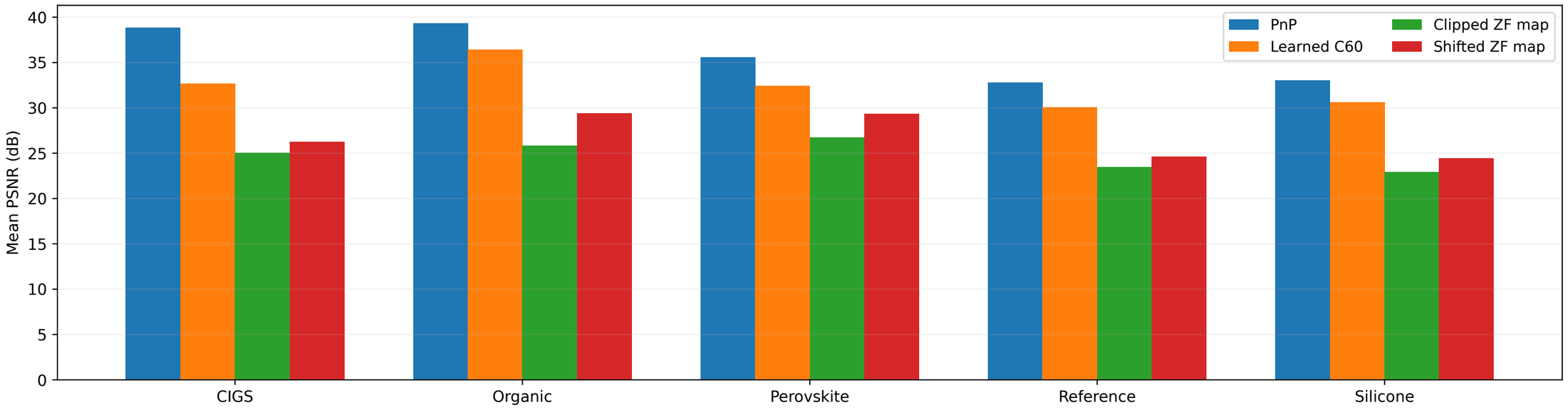
10 frozen PCM validation images (2/category) | 20% Hadamard | NPL-default noise | 512 PDHG iterations

Mean PSNR (dB): PnP 35.896 | Learned C60 32.427 | Clipped ZF 24.790 | Shifted ZF 26.800

Shifted versus clipped: +2.010 dB; shifted versus learned: -5.627 dB.

Shifted beats clipped on 10/10 images; shifted beats learned on 0/10. Best rho at lower boundary 0/10; outside final bracket 1/10.

New positive map: $\Lambda_\rho(x) = \rho[x_{\mathrm{ZF}}(x) - \min(x_{\mathrm{ZF}}) + 10^{-5}]$. No clipping; same image shape and contrast.



Interpretation

This repeat retains signed zero-filled pixel contrast by adding a constant offset to every pixel, so the minimum equals 1e-5 before rho scaling. All experiments use the same archived measurements and unchanged C60 checkpoint and PDHG. Rho was selected using GT PSNR on each validation image. The clipped and shifted results both use finite search ranges; boundary results and non-unimodality must be considered before drawing a general conclusion.

The validation subset and PnP baselines are identical to the earlier clipped zero-fill probe.

Per-image scores and parameter multipliers

Positive PSNR gain means shifted zero-fill outperforms the previous clipped zero-fill map.

Image	Old rho	Shift rho	PnP PSNR	Learned PSNR	Clipped PSNR	Shifted PSNR	Shift - clip	Shift - learned
CIGS_0768	0.0001	0.10347	37.39	32.22	24.34	25.28	+0.95	-6.93
CIGS_0158	0.0001	0.096441	40.24	33.04	25.76	27.22	+1.46	-5.82
Organic_0917	0.0001	0.083358	39.56	36.50	25.91	29.81	+3.89	-6.70
Organic_0008	0.0001	0.087719	39.13	36.31	25.77	29.01	+3.24	-7.30
Perovskite_0733	0.0001	0.14776	38.56	36.29	28.71	31.31	+2.61	-4.98
Perovskite_0396	0.0001	0.097974	32.52	28.56	24.71	27.31	+2.60	-1.25
Reference_0768	0.0001	0.12857	33.48	31.75	25.09	26.40	+1.32	-5.35
Reference_0589	0.0001	0.096282	32.07	28.39	21.81	22.76	+0.94	-5.63
Silicone_0008	0.001	0.096441	31.87	29.79	23.25	26.09	+2.84	-3.70
Silicone_0589	0.0001	0.001	34.15	31.43	22.55	22.81	+0.26	-8.61

Search protocol and cautions

Fixed rho interval [0.0001,10]. Linear-rho ternary search stops at interval width 0.01. Log-spaced diagnostic candidates and final bracket probes are also evaluated.

The best observed candidate is selected, even if it is outside the final ternary interval. Non-unimodality is possible; this is not a mathematically certified global optimum.

The learned checkpoint and PnP hyperparameters were selected using validation data before this experiment. The rho multiplier is also chosen with validation GT, so all conclusions are exploratory.

Same 10 frozen validation acquisitions, same C60 and PnP baselines. Per-image search traces are archived.

Experiment definition and positivity treatment

Two simple maps from exactly the same zero-filled reconstruction, compared with learned C60 and PnP.

Shared solver

C60 epoch-60 checkpoint; 512 TV-PDHG iterations; 20 CG iterations for original learned-map input; zero image initialization; normalized 20% multilevel Hadamard forward operator. No LION source changes.

Previous clipped map

$\Lambda_\rho^{\text{clip}}(x) = \rho \max(x_{\text{ZF}}(x), 0)$. Negative pixels were set to zero; this changes the local contrast and may give exactly zero weights.

New shifted map

$\Lambda_\rho^{\text{shift}}(x) = \rho[x_{\text{ZF}}(x) - \min(x_{\text{ZF}}) + 10^{-5}]$. Uses each image's own raw minimum. No clipping, no upper bound. The parameter map minimum BEFORE rho is 1e-5.

Original learned map

The original learned positive C60 parameter map is predicted by the same pretrained CNN. It is neither shifted nor globally rescaled; it is recomputed and verified against the earlier archived C60 output.

Comparison

GT, zero-fill, PnP-ADMM-tuned-5 (rho=0.5, sigma=0.075), C60 learned-map, C60 clipped-map with best observed rho, and C60 shifted-map with best observed rho.

Selection and interpretation

One rho per image chosen by GT PSNR. C60 learned-map and PnP baseline are fixed. The scalar selection is oracle-only and may not generalise; repeat experiments on independent data are needed for claims about performance.

Full acquisitions, individual map arrays and each rho-evaluation trace are in the complete ZIP archive.

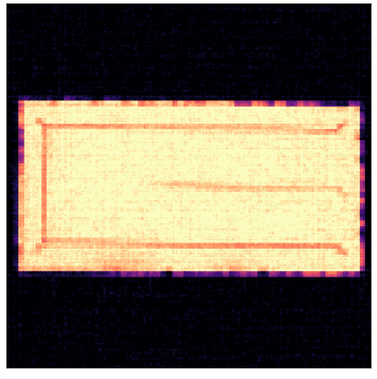
01. CIGS_0768

Validation position 907 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

Ground truth



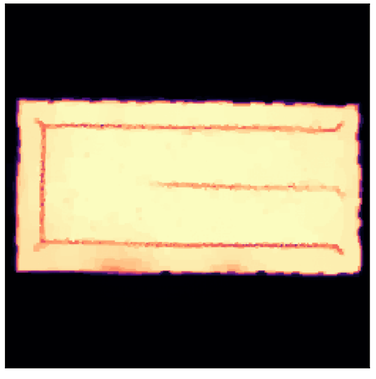
Zero-filled
PSNR 24.19 dB | SSIM 0.436
HaarPSI 0.460



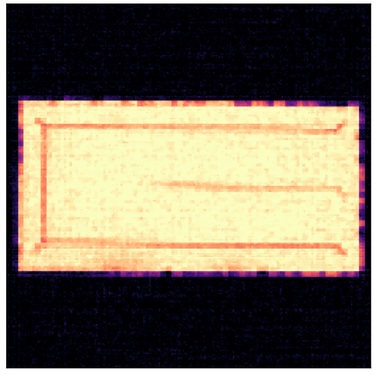
PnP-ADMM-tuned-5
PSNR 37.39 dB | SSIM 0.984
HaarPSI 0.943



Learned C60
PSNR 32.22 dB | SSIM 0.954
HaarPSI 0.869



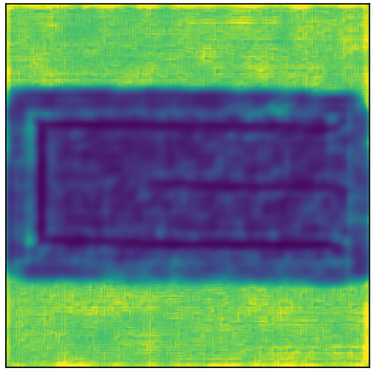
Clipped ZF | rho=0.0001
PSNR 24.34 dB | SSIM 0.450
HaarPSI 0.482



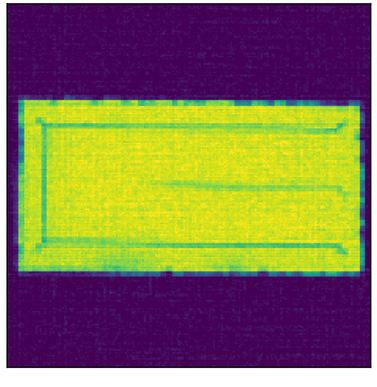
Shifted ZF | rho=0.1035
PSNR 25.28 dB | SSIM 0.619
HaarPSI 0.563



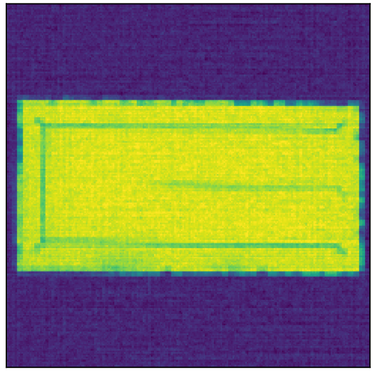
Learned lambda map
min 0.00399 | mean 0.113 | max 0.286



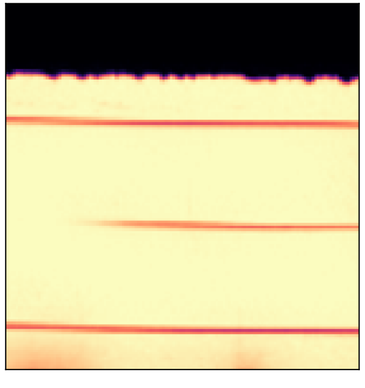
Clipped ZF lambda * rho
min 0 | mean 4.33e-05 | max 0.000113



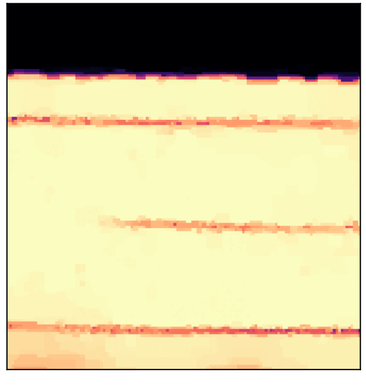
Shifted ZF lambda * rho
min 1.03e-06 | mean 0.0574 | max 0.13



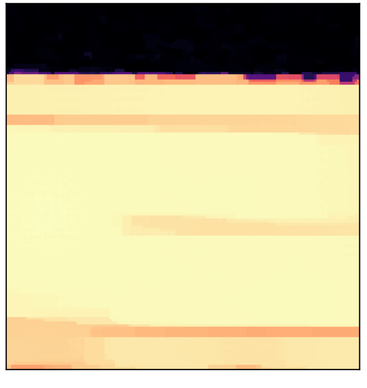
GT crop



Learned C60 crop



Shifted ZF crop



Shifted vs clipped: +0.947 dB | Shifted vs learned: -6.934 dB

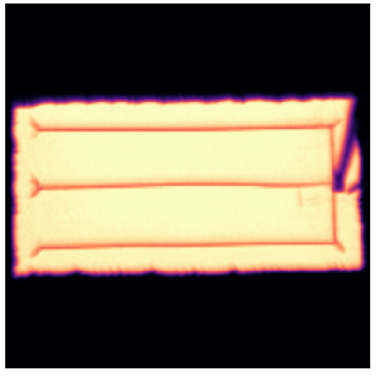
Shifted rho evaluations: 49, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.12831. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

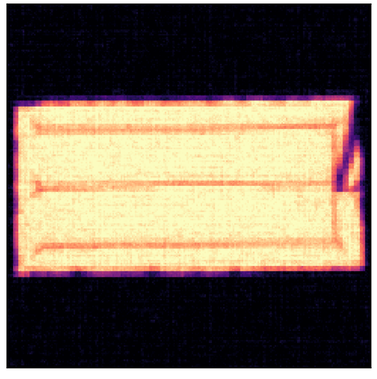
02. CIGS_0158

Validation position 910 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

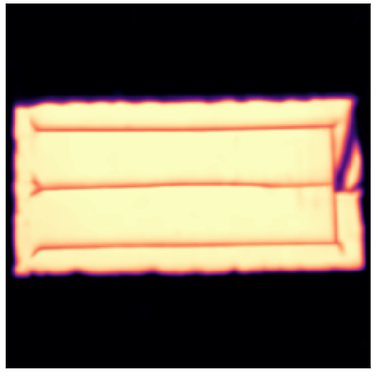
Ground truth



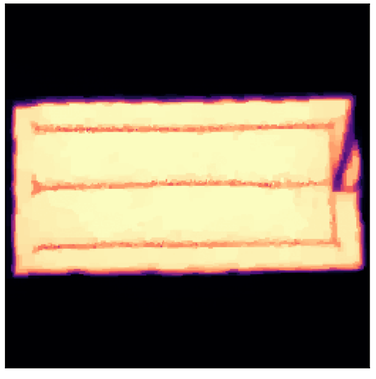
Zero-filled
PSNR 25.69 dB | SSIM 0.483
HaarPSI 0.535



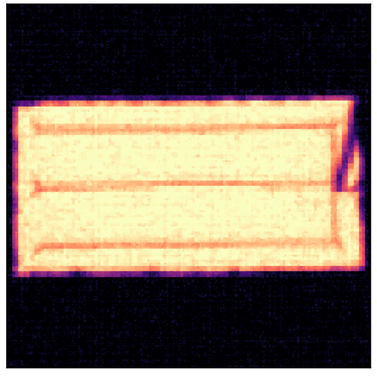
PnP-ADMM-tuned-5
PSNR 40.24 dB | SSIM 0.984
HaarPSI 0.952



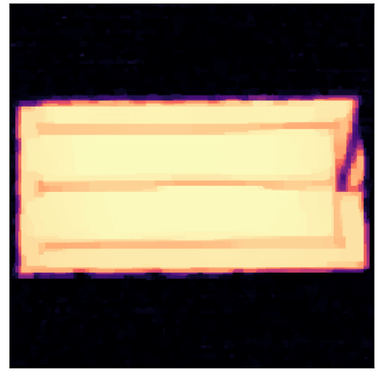
Learned C60
PSNR 33.04 dB | SSIM 0.944
HaarPSI 0.825



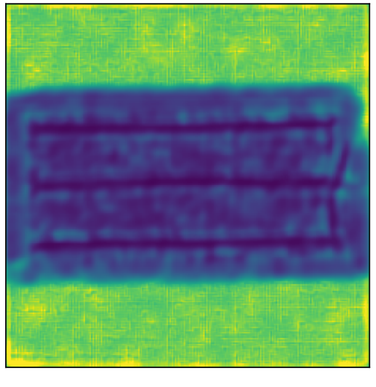
Clipped ZF | rho=0.0001
PSNR 25.76 dB | SSIM 0.495
HaarPSI 0.558



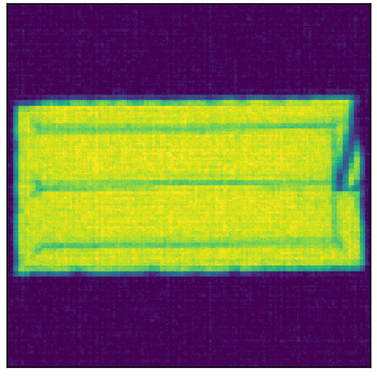
Shifted ZF | rho=0.09644
PSNR 27.22 dB | SSIM 0.684
HaarPSI 0.618



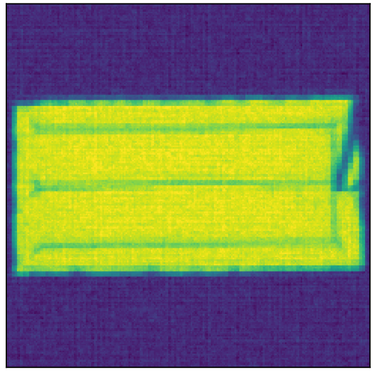
Learned lambda map
min 0.00461 | mean 0.112 | max 0.31



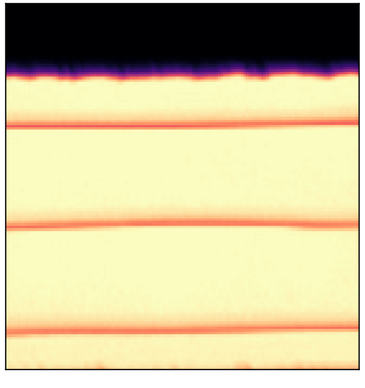
Clipped ZF lambda * rho
min 0 | mean 4.31e-05 | max 0.000112



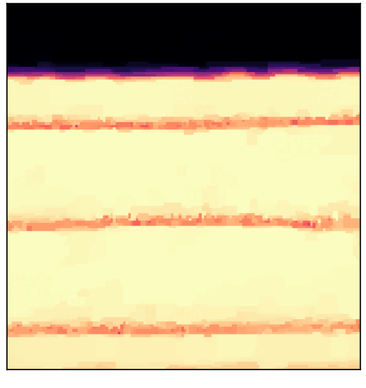
Shifted ZF lambda * rho
min 9.64e-07 | mean 0.0552 | max 0.122



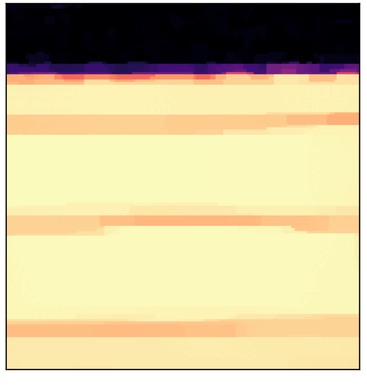
GT crop



Learned C60 crop



Shifted ZF crop



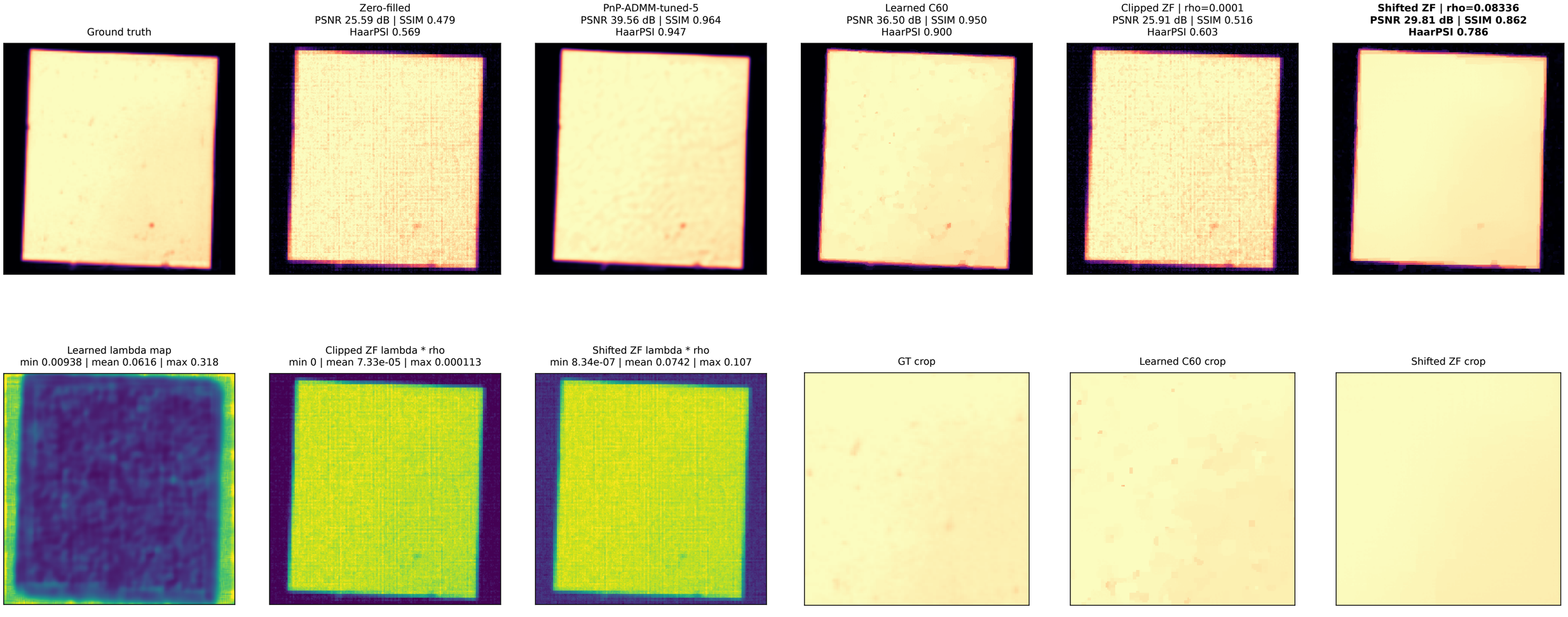
Shifted vs clipped: +1.459 dB | Shifted vs learned: -5.821 dB

Shifted rho evaluations: 49, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.14823. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

03. Organic_0917

Validation position 902 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho



Shifted vs clipped: +3.892 dB | Shifted vs learned: -6.697 dB

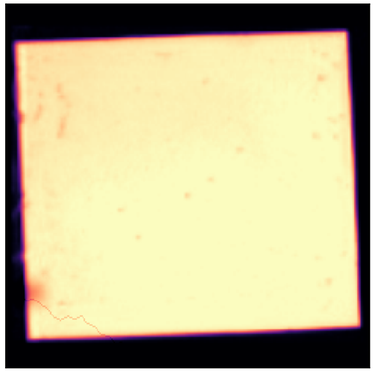
Shifted rho evaluations: 48, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.15938. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

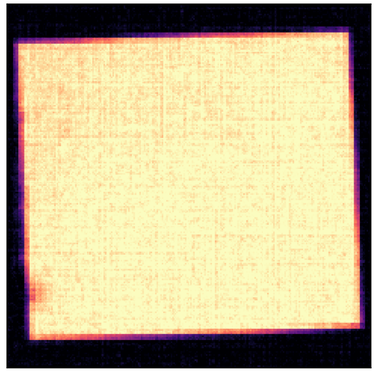
04. Organic_0008

Validation position 903 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

Ground truth



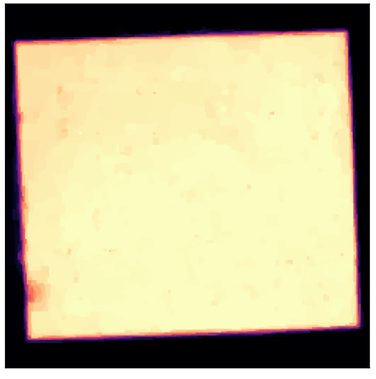
Zero-filled
PSNR 25.45 dB | SSIM 0.472
HaarPSI 0.545



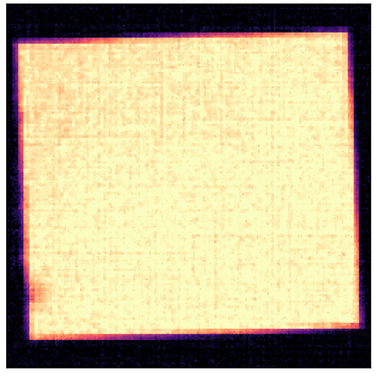
PnP-ADMM-tuned-5
PSNR 39.13 dB | SSIM 0.965
HaarPSI 0.947



Learned C60
PSNR 36.31 dB | SSIM 0.948
HaarPSI 0.888



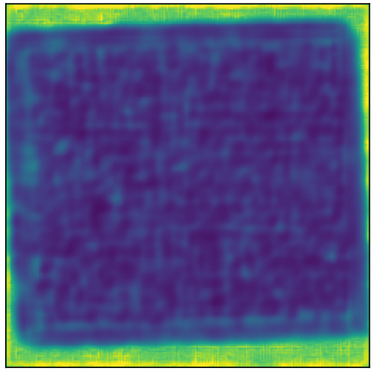
Clipped ZF | rho=0.0001
PSNR 25.77 dB | SSIM 0.509
HaarPSI 0.580



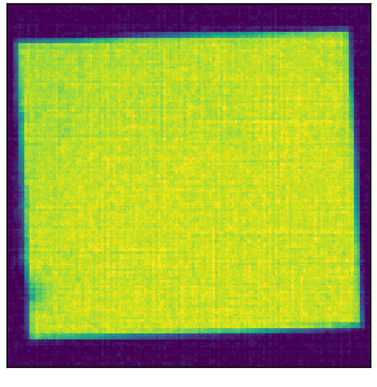
Shifted ZF | rho=0.08772
PSNR 29.01 dB | SSIM 0.852
HaarPSI 0.767



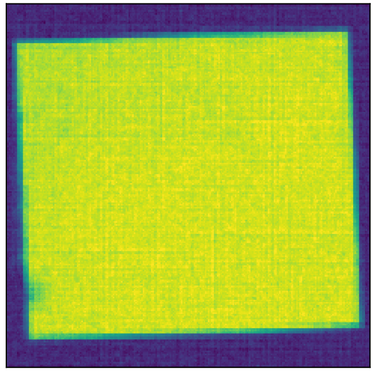
Learned lambda map
min 0.0107 | mean 0.0625 | max 0.301



Clipped ZF lambda * rho
min 0 | mean 7.4e-05 | max 0.000114



Shifted ZF lambda * rho
min 8.77e-07 | mean 0.0773 | max 0.112



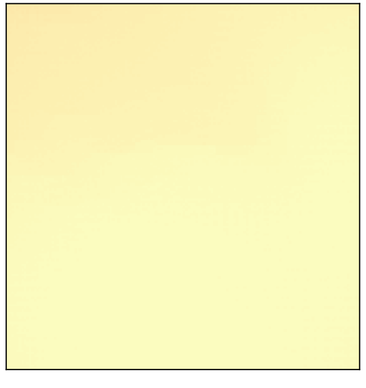
GT crop



Learned C60 crop



Shifted ZF crop



Shifted vs clipped: +3.238 dB | Shifted vs learned: -7.296 dB

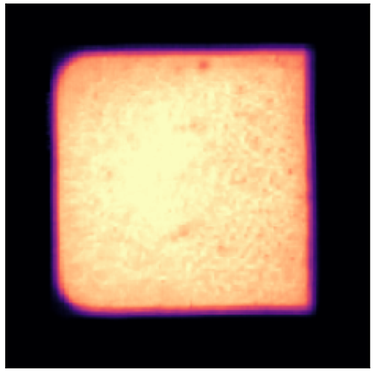
Shifted rho evaluations: 49, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.14407. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

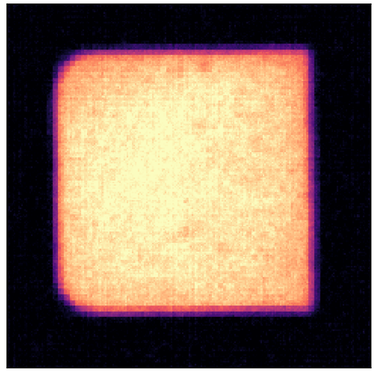
05. Perovskite_0733

Validation position 901 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

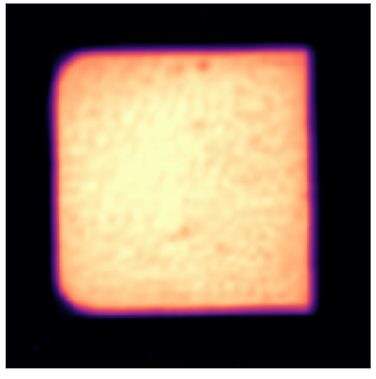
Ground truth



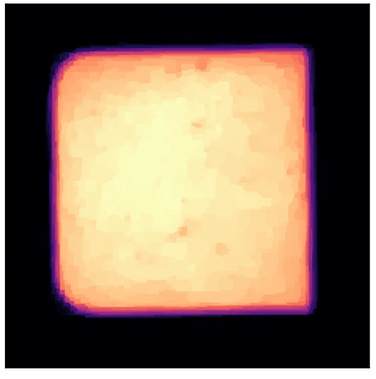
Zero-filled
PSNR 28.88 dB | SSIM 0.543
HaarPSI 0.685



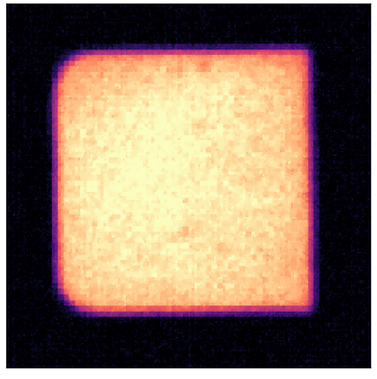
PnP-ADMM-tuned-5
PSNR 38.56 dB | SSIM 0.946
HaarPSI 0.922



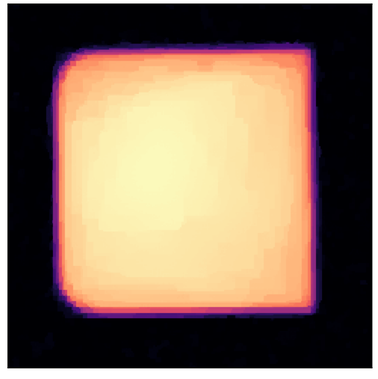
Learned C60
PSNR 36.29 dB | SSIM 0.890
HaarPSI 0.854



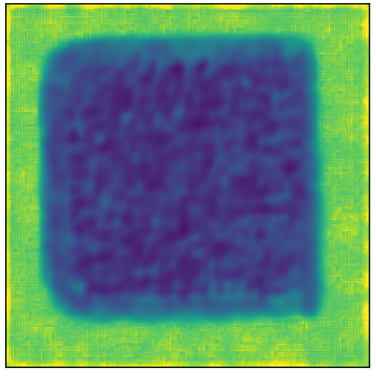
Clipped ZF | rho=0.0001
PSNR 28.71 dB | SSIM 0.559
HaarPSI 0.706



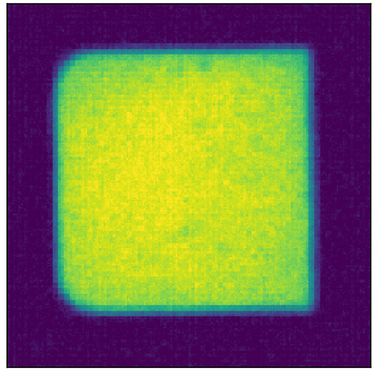
Shifted ZF | rho=0.1478
PSNR 31.31 dB | SSIM 0.785
HaarPSI 0.786



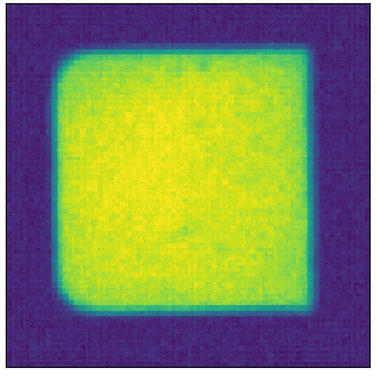
Learned lambda map
min 0.0138 | mean 0.107 | max 0.294



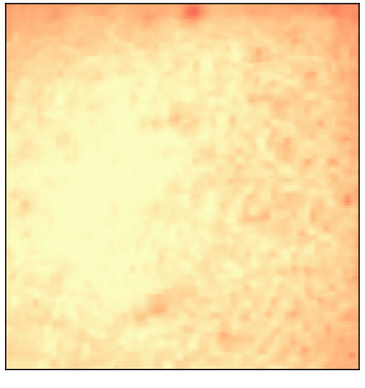
Clipped ZF lambda * rho
min 0 | mean 4.63e-05 | max 0.00011



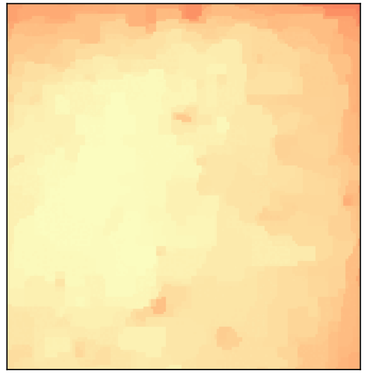
Shifted ZF lambda * rho
min 1.48e-06 | mean 0.0863 | max 0.181



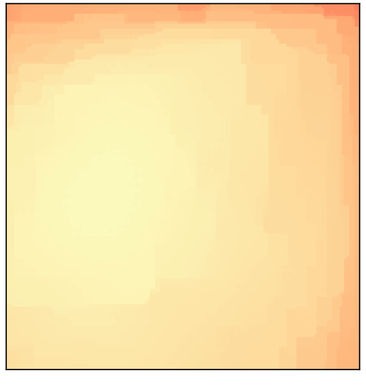
GT crop



Learned C60 crop



Shifted ZF crop



Shifted vs clipped: +2.607 dB | Shifted vs learned: -4.978 dB

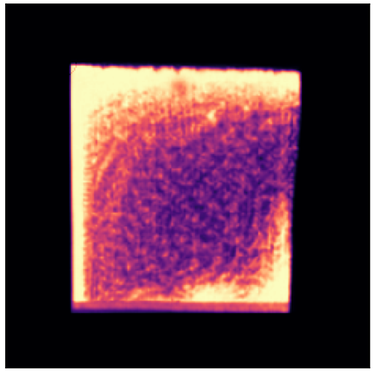
Shifted rho evaluations: 48, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.12647. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

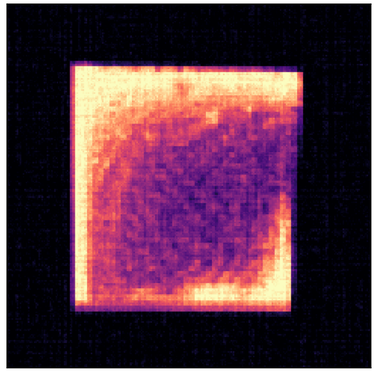
06. Perovskite_0396

Validation position 906 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

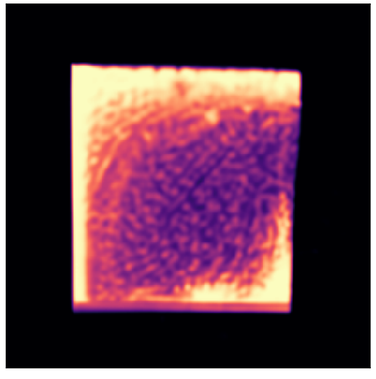
Ground truth



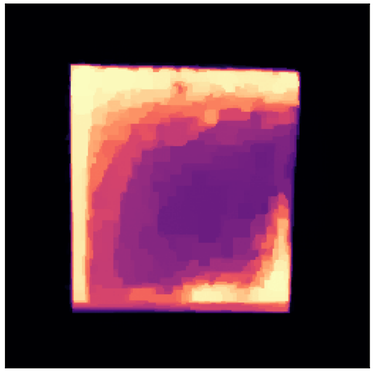
Zero-filled
PSNR 24.60 dB | SSIM 0.473
HaarPSI 0.515



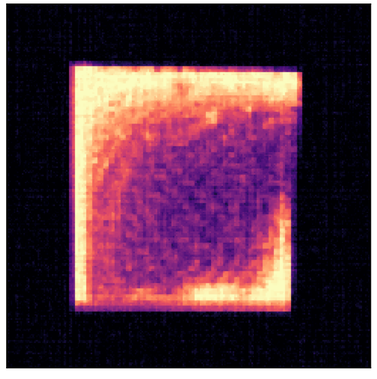
PnP-ADMM-tuned-5
PSNR 32.52 dB | SSIM 0.931
HaarPSI 0.818



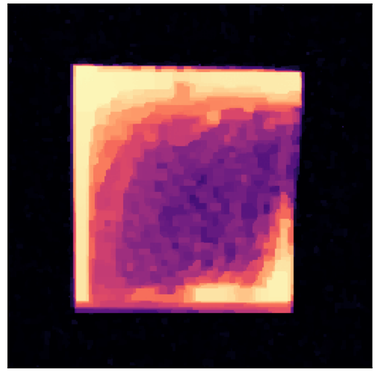
Learned C60
PSNR 28.56 dB | SSIM 0.803
HaarPSI 0.552



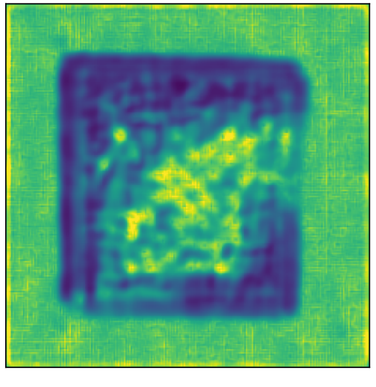
Clipped ZF | rho=0.0001
PSNR 24.71 dB | SSIM 0.467
HaarPSI 0.524



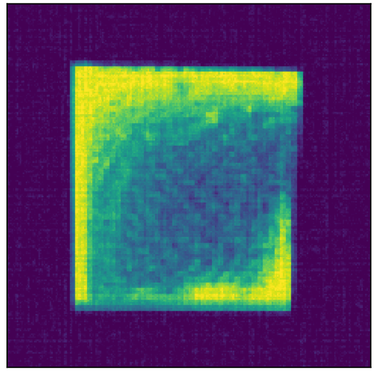
Shifted ZF | rho=0.09797
PSNR 27.31 dB | SSIM 0.606
HaarPSI 0.601



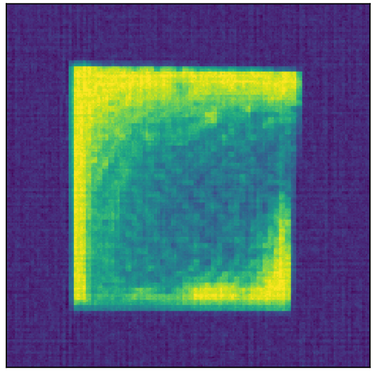
Learned lambda map
min 0.00891 | mean 0.15 | max 0.349



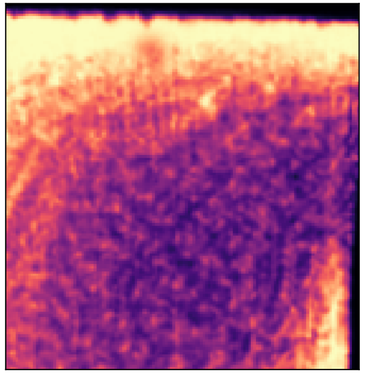
Clipped ZF lambda * rho
min 0 | mean 2.57e-05 | max 0.000112



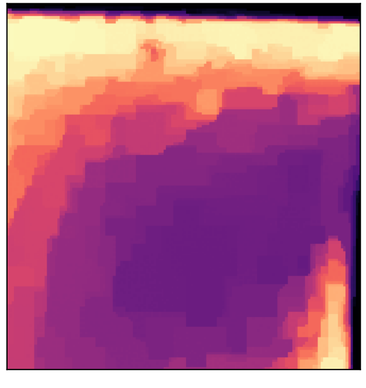
Shifted ZF lambda * rho
min 9.8e-07 | mean 0.0386 | max 0.124



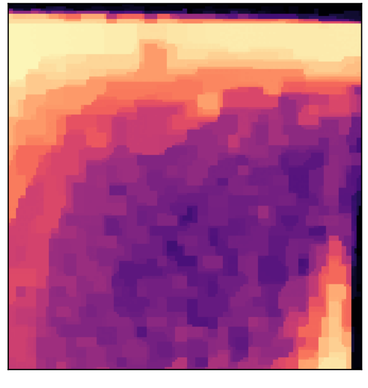
GT crop



Learned C60 crop



Shifted ZF crop



Shifted vs clipped: +2.602 dB | Shifted vs learned: -1.248 dB

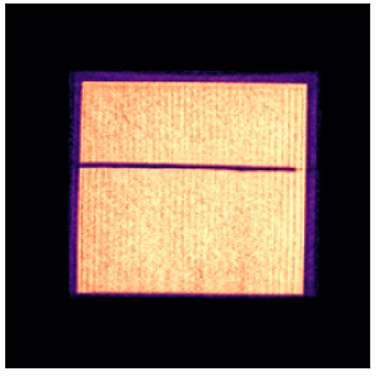
Shifted rho evaluations: 49, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.14521. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

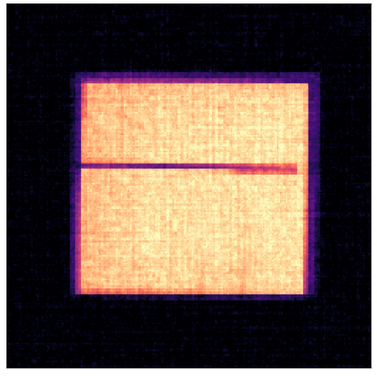
07. Reference_0768

Validation position 907 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

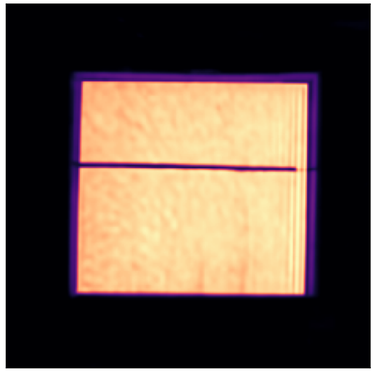
Ground truth



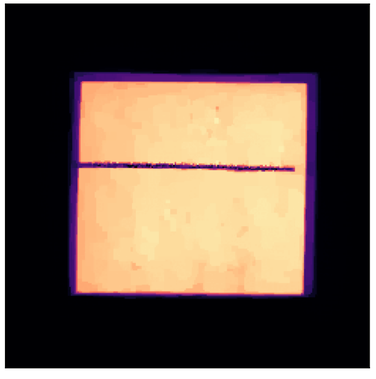
Zero-filled
PSNR 25.09 dB | SSIM 0.436
HaarPSI 0.531



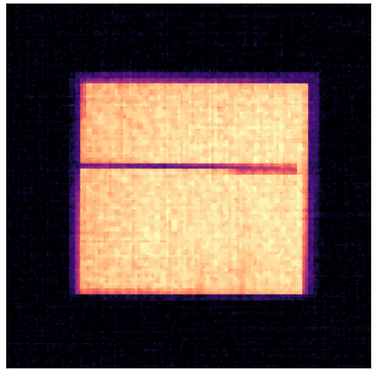
PnP-ADMM-tuned-5
PSNR 33.48 dB | SSIM 0.860
HaarPSI 0.789



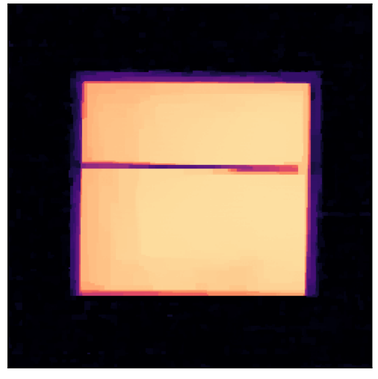
Learned C60
PSNR 31.75 dB | SSIM 0.807
HaarPSI 0.666



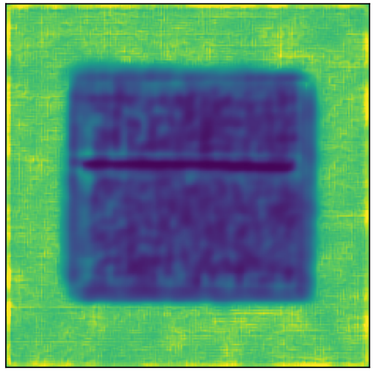
Clipped ZF | rho=0.0001
PSNR 25.09 dB | SSIM 0.420
HaarPSI 0.537



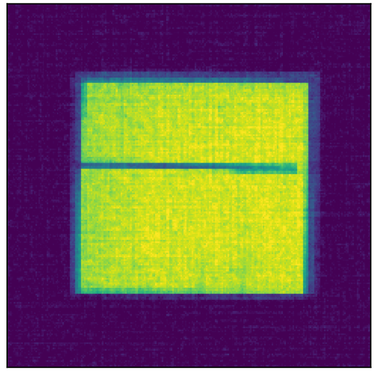
Shifted ZF | rho=0.1286
PSNR 26.40 dB | SSIM 0.566
HaarPSI 0.554



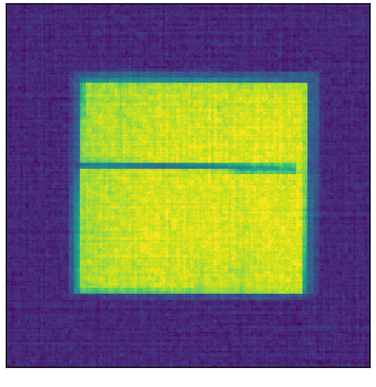
Learned lambda map
min 0.00289 | mean 0.128 | max 0.319



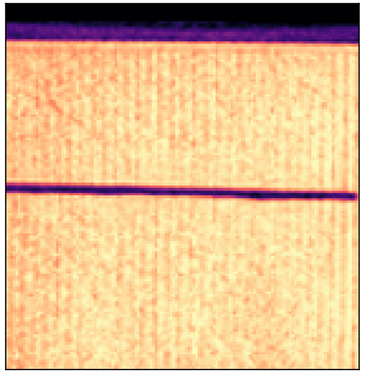
Clipped ZF lambda * rho
min 0 | mean 3.44e-05 | max 0.000106



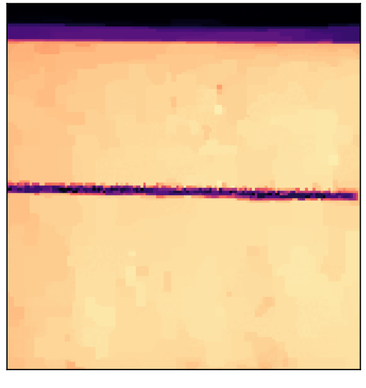
Shifted ZF lambda * rho
min 1.29e-06 | mean 0.0599 | max 0.153



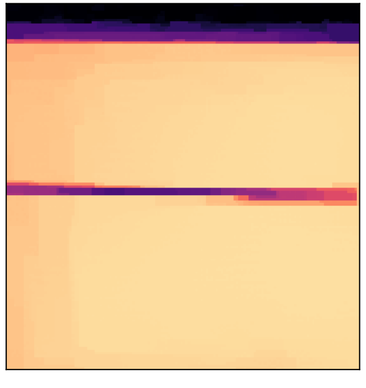
GT crop



Learned C60 crop



Shifted ZF crop



Shifted vs clipped: +1.316 dB | Shifted vs learned: -5.349 dB

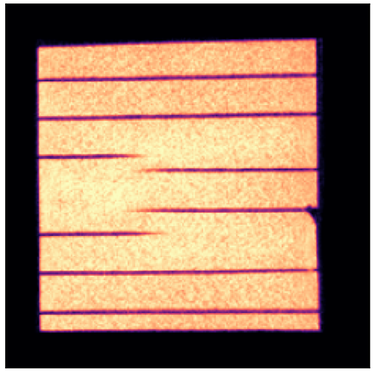
Shifted rho evaluations: 48, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.12983. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

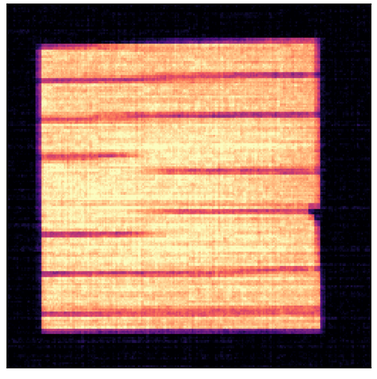
08. Reference_0589

Validation position 909 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

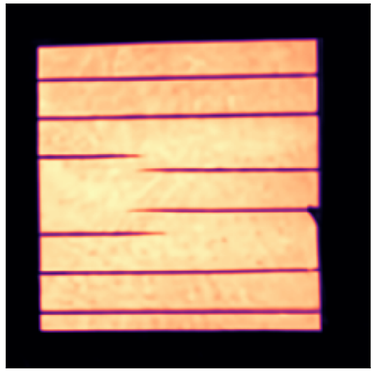
Ground truth



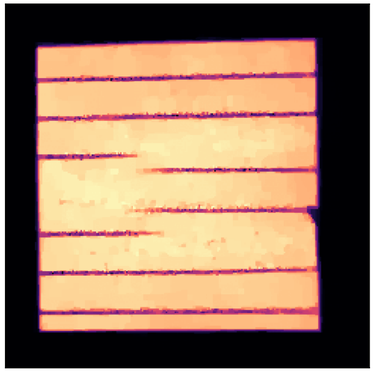
Zero-filled
PSNR 21.63 dB | SSIM 0.422
HaarPSI 0.484



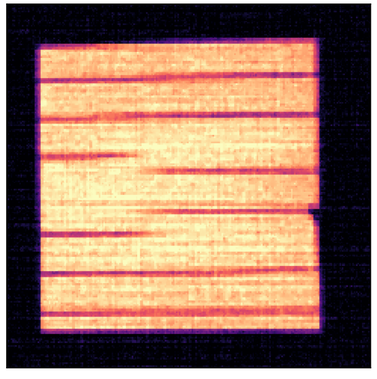
PnP-ADMM-tuned-5
PSNR 32.07 dB | SSIM 0.840
HaarPSI 0.799



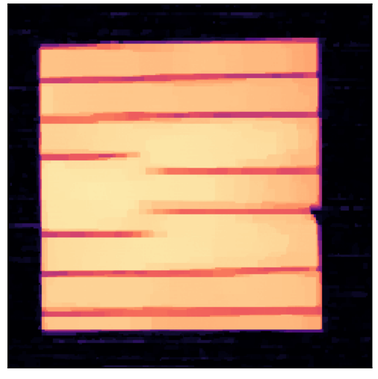
Learned C60
PSNR 28.39 dB | SSIM 0.798
HaarPSI 0.723



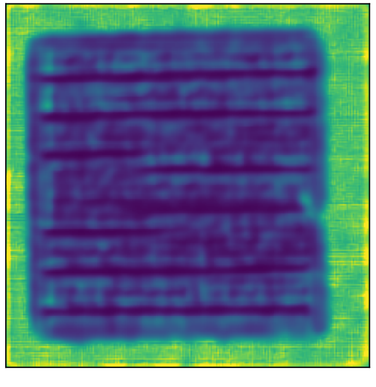
Clipped ZF | rho=0.0001
PSNR 21.81 dB | SSIM 0.426
HaarPSI 0.496



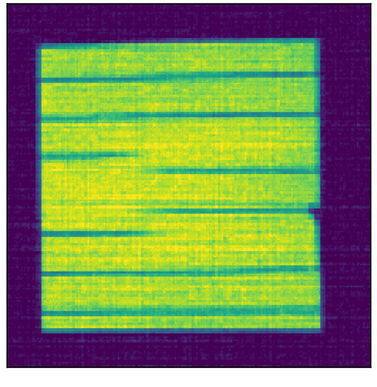
Shifted ZF | rho=0.09628
PSNR 22.76 dB | SSIM 0.531
HaarPSI 0.488



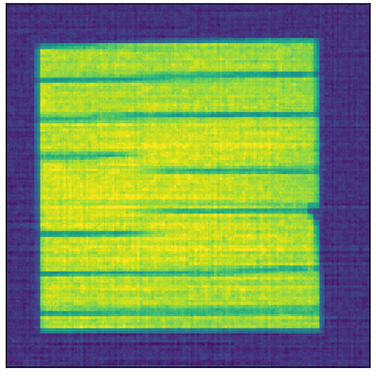
Learned lambda map
min 0.00223 | mean 0.0913 | max 0.365



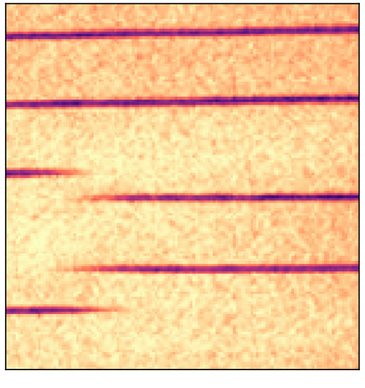
Clipped ZF lambda * rho
min 0 | mean 5.3e-05 | max 0.000115



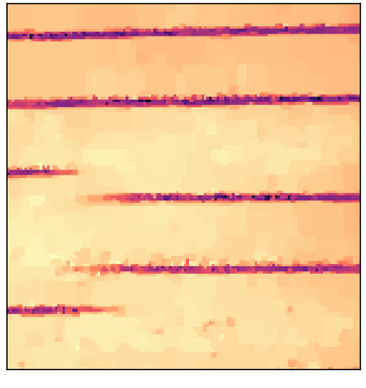
Shifted ZF lambda * rho
min 9.63e-07 | mean 0.0677 | max 0.128



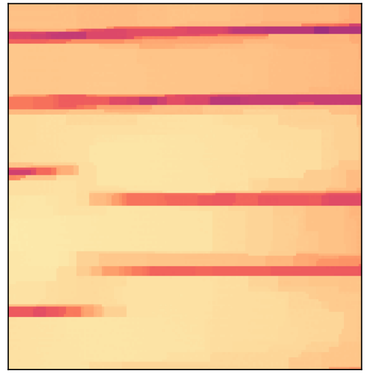
GT crop



Learned C60 crop



Shifted ZF crop



Shifted vs clipped: +0.943 dB | Shifted vs learned: -5.632 dB

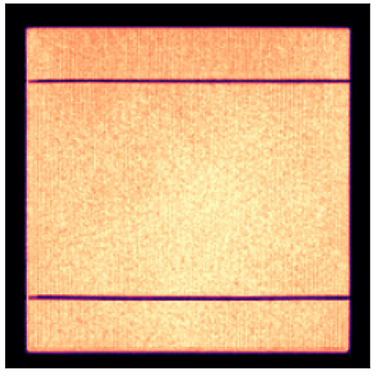
Shifted rho evaluations: 49, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.17928. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

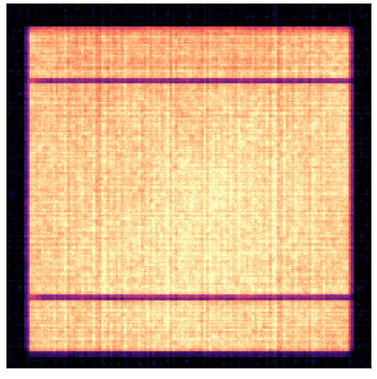
09. Silicone_0008

Validation position 903 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho

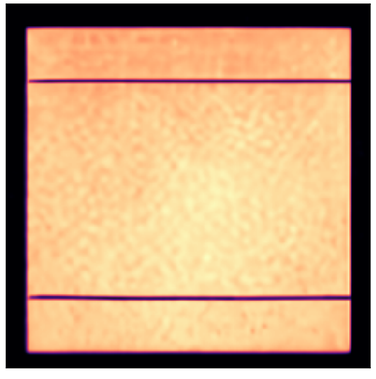
Ground truth



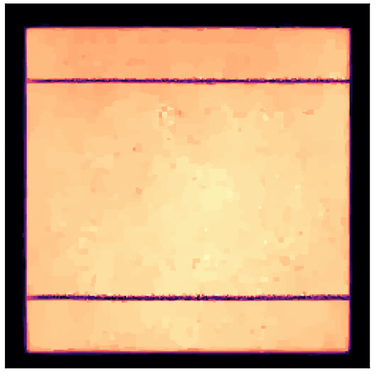
Zero-filled
PSNR 22.69 dB | SSIM 0.427
HaarPSI 0.533



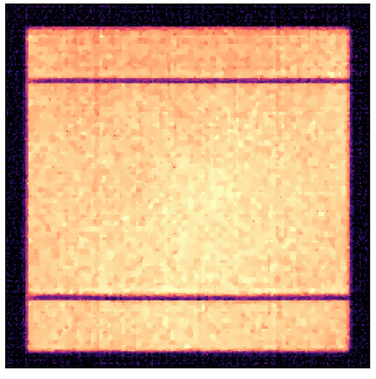
PnP-ADMM-tuned-5
PSNR 31.87 dB | SSIM 0.743
HaarPSI 0.843



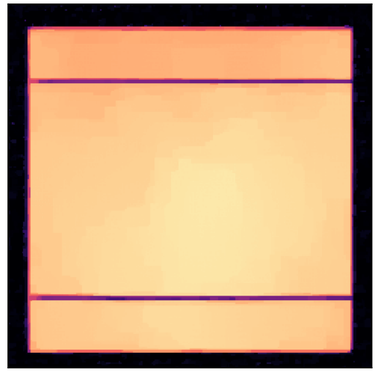
Learned C60
PSNR 29.79 dB | SSIM 0.718
HaarPSI 0.803



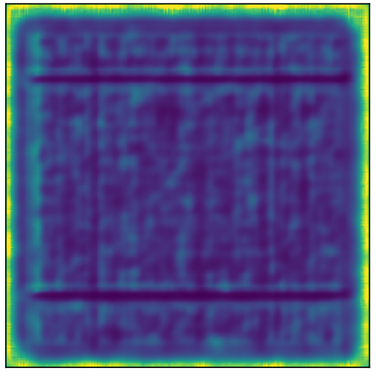
Clipped ZF | rho=0.001
PSNR 23.25 dB | SSIM 0.533
HaarPSI 0.574



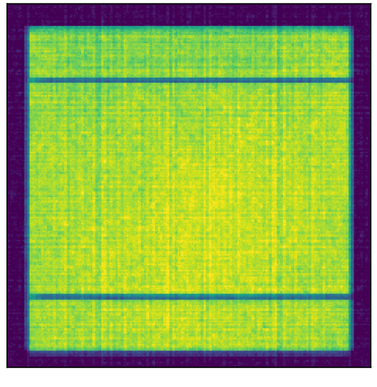
Shifted ZF | rho=0.09644
PSNR 26.09 dB | SSIM 0.599
HaarPSI 0.654



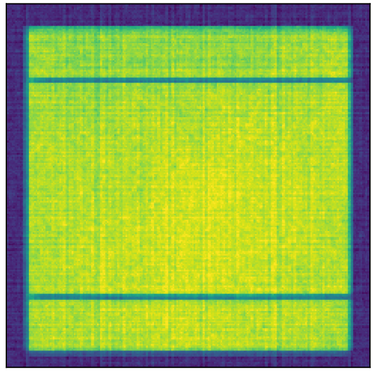
Learned lambda map
min 0.00184 | mean 0.0614 | max 0.379



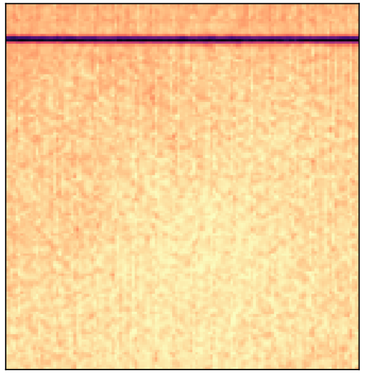
Clipped ZF lambda * rho
min 0 | mean 0.000703 | max 0.00113



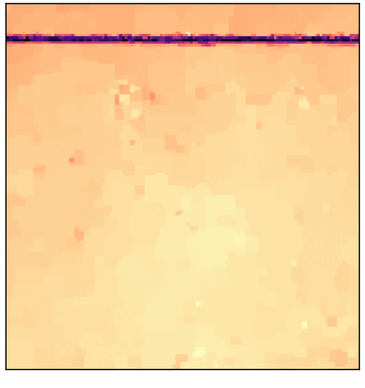
Shifted ZF lambda * rho
min 9.64e-07 | mean 0.0819 | max 0.123



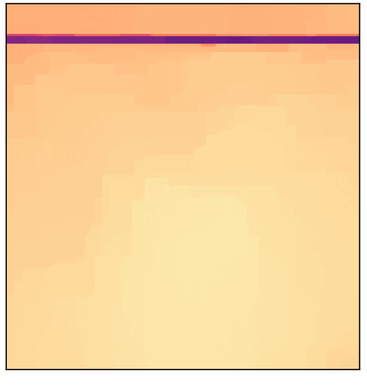
GT crop



Learned C60 crop



Shifted ZF crop



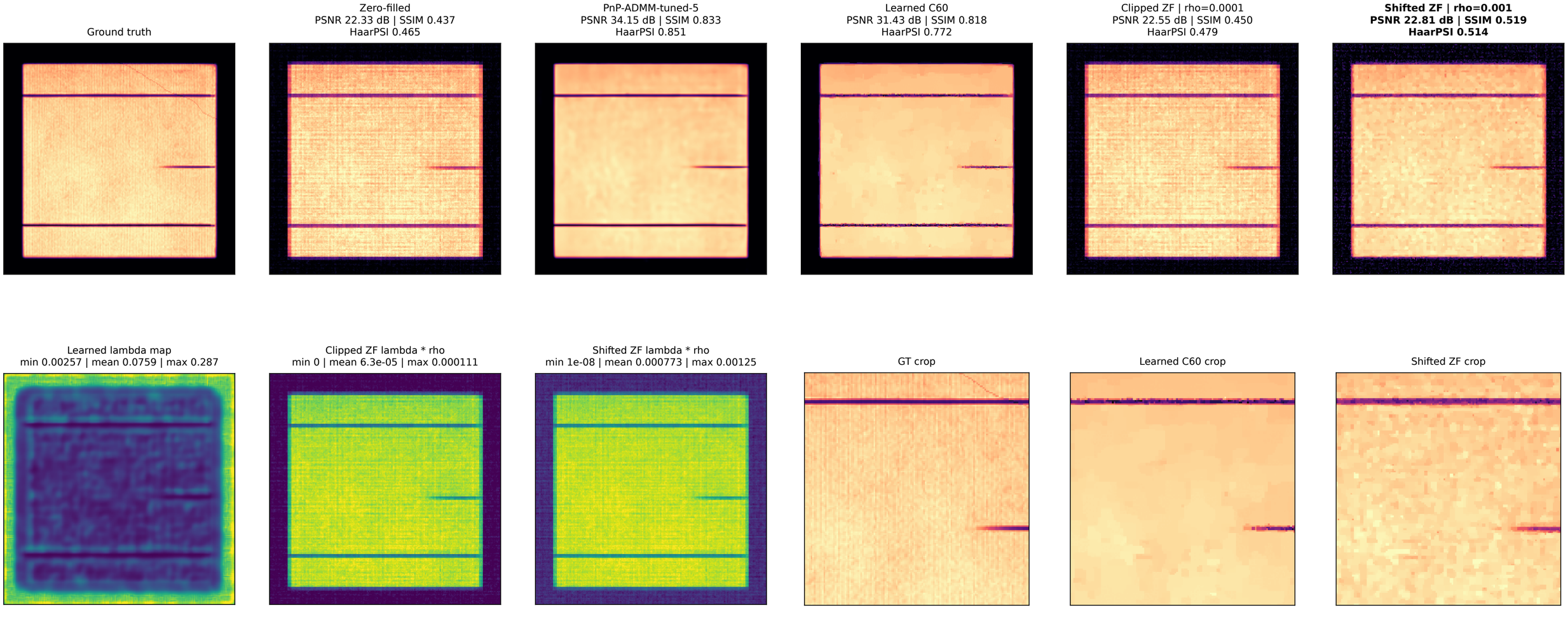
Shifted vs clipped: +2.839 dB | Shifted vs learned: -3.703 dB

Shifted rho evaluations: 49, final width 0.00677, best within final bracket. Zero-fill shift offset: 0.14922. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.

10. Silicone_0589

Validation position 909 | same measurement | same C60 (512 iterations) | per-image GT-PSNR selected rho



Shifted vs clipped: +0.261 dB | Shifted vs learned: -8.615 dB

Shifted rho evaluations: 48, final width 0.00677, best outside final bracket. Zero-fill shift offset: 0.14644. Image panels share [0,1]; parameter maps have independent colour scales.

All reconstructions and GT are raw and not clipped when computing metrics. Orange/dark image colours are for display only.